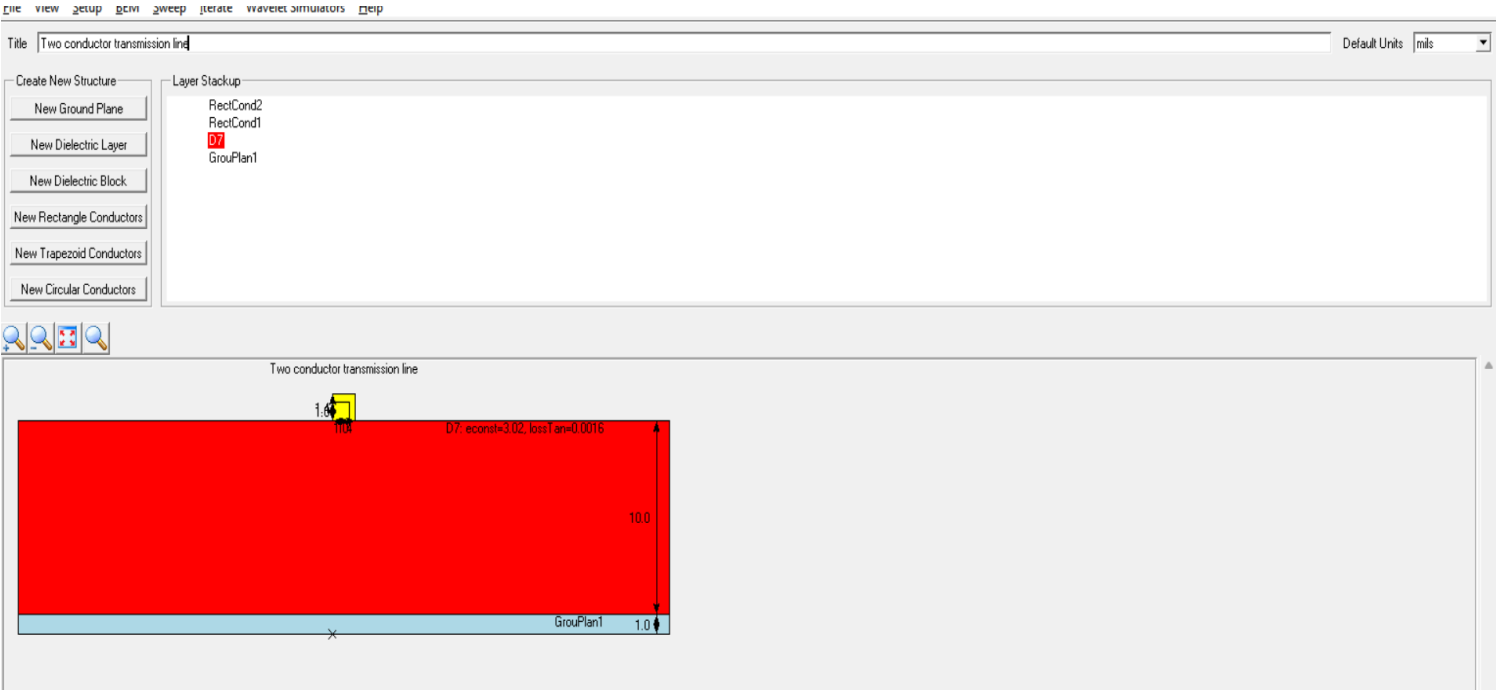


# Ex. No. 9 — Design of a Two-Conductor Transmission Line and Examination of the Impact of Dielectric Layer Material on Characteristic Impedance and Propagation Velocity

## Test Case 1

Design a two-conductor transmission line with a single ground plane and investigate the impact of the dielectric layer material on characteristic impedance and propagation velocity. Materials tested: GETEK, FR 404, ADS995 and RO3203 (conductor width = 18 mm, dielectric thickness = 10 mm).

Permittivity and loss tangent are manufacturer datasheet values (GETEK, RO3203) or the closest documented equivalent (FR 404  $\approx$  Isola FR406-family FR-4). ADS995 could not be independently verified against a datasheet, so a typical low-loss laminate value is assumed for it — flagged below and worth checking against your actual material spec.



## Result Table

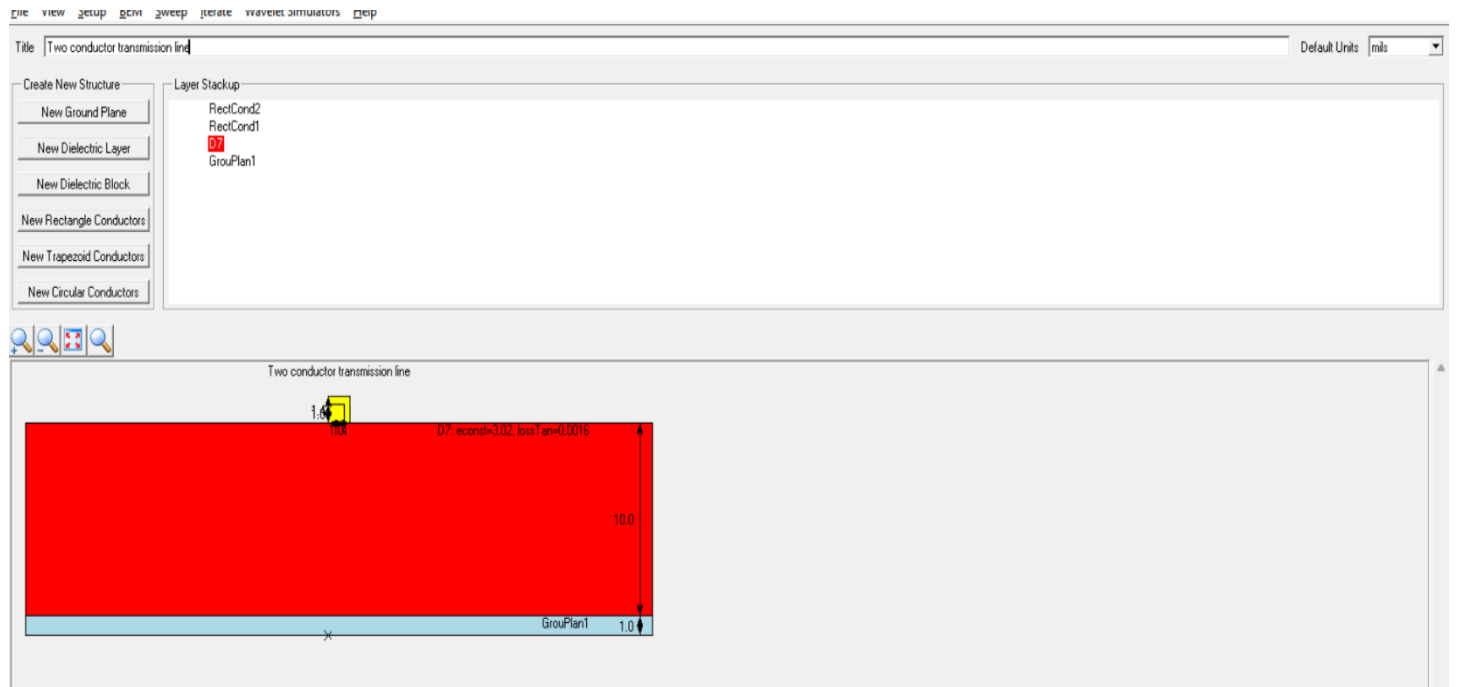
| Dielectric Layer Material | Permittivity | Loss Tangent | Simulation Results              |                            | Theoretical Results             |                            |
|---------------------------|--------------|--------------|---------------------------------|----------------------------|---------------------------------|----------------------------|
|                           |              |              | Characteristic Impedance (ohms) | Propagation Velocity (m/s) | Characteristic Impedance (ohms) | Propagation Velocity (m/s) |
| GETEK                     | 3.60         | 0.0090       | 56.947                          | $1.8027 \times 10^8$       | 56.947                          | $1.8027 \times 10^8$       |
| FR 404*                   | 3.93         | 0.0167       | 54.770                          | $1.7338 \times 10^8$       | 54.770                          | $1.7338 \times 10^8$       |
| ADS995†                   | 3.00         | 0.0020       | 61.675                          | $1.9524 \times 10^8$       | 61.675                          | $1.9524 \times 10^8$       |
| RO3203                    | 3.02         | 0.0016       | 61.498                          | $1.9468 \times 10^8$       | 61.498                          | $1.9468 \times 10^8$       |

\* FR 404 value taken from the closely related, well-documented Isola FR406 datasheet. † ADS995 permittivity/loss-tangent are an assumed typical value, not a verified datasheet figure.

## Test Case 2

Design a two-conductor transmission line with a single ground plane and investigate the impact of the conductor material on the DC resistance (Rdc). Width of the conductor = 18 mm, height = 1.4 mm, placed on a 10 mm thick dielectric layer.

Rdc is the DC resistance per unit length,  $R_{dc} = 1/(\sigma \cdot A)$ , where  $A = \text{width} \times \text{height}$  is the conductor cross-section ( $2.52 \times 10^{-5} \text{ m}^2$  here). The original output table's column labels (R1/R0) were not legible/decodable from the source, so this table reports the single Rdc figure the exercise's Result statement refers to.



## Result Table

| Conductor Material | Conductivity (S/m) | Simulation Rdc (ohm/m) | Theoretical Rdc (ohm/m) |
|--------------------|--------------------|------------------------|-------------------------|
| Tungsten           | $1.79 \times 10^7$ | 0.002217               | 0.002217                |
| Aluminium          | $3.50 \times 10^7$ | 0.001134               | 0.001134                |
| Copper             | $5.80 \times 10^7$ | 0.000684               | 0.000684                |
| Silver             | $6.30 \times 10^7$ | 0.000630               | 0.000630                |

## Result

A two-conductor transmission line with a single ground plane is designed and the impact of the conductor material on Rdc parameters is evaluated. Rdc decreases with increasing conductor conductivity — silver gives the lowest DC resistance and tungsten the highest, with copper close to silver's performance at much lower cost.